

From messy data to clear answers — a 30-second look

What "data engineering" actually means, shown on two real rows from the project.

BEFORE — the raw data, exactly as it arrives

Car	Price	Power	Engine	Fuel
Ferrari SF90 Stradale	\$1,100,000	963 hp	3990 cc	plug in hyrbrid
Ford KA+	\$12,000-\$15,000	70-85 hp	1,200 cc	Petrol

A computer chokes on all of this: prices are **text** with dollar signs, commas, even **ranges** (\$12,000–\$15,000); power and engine size are **buried inside text** ("963 hp", "3990 cc"); and there's a **typo** — "plug in hyrbrid." You can't sort, add up or chart any of it as-is.

↓ **the pipeline cleans, types and enriches every value** ↓

AFTER — clean, computer-ready data (same two cars)

Car	Price (USD)	Power (hp)	Engine (cc)	Fuel	Segment
Ferrari SF90 Stradale	1,100,000	963	3,990	Hybrid	Luxury
Ford KA+	12,000	70	1,200	Petrol	Economy

Same cars, now usable: every value is a **proper number** the computer can add and compare, the typo is fixed to a real category (**Hybrid**), and useful new labels are added — like the **price Segment** that groups cars into Economy / Premium / Luxury.

SO WHAT — now the data can answer real questions

Once thousands of rows are clean like this, you can ask things a spreadsheet never could. One example from this project: **luxury cars cost about 4× more per unit of power** than ordinary cars (\$714 vs ~\$170 per horsepower). Above a point, you're paying for the brand and the prestige — not the engine.

Real rows from the project dataset (Ferrari SF90 Stradale, Ford KA+), shown before and after the cleaning pipeline. Full findings in [dashboard-insights.pdf](#).